TEAVISION360 – A COMPLETE AI AND IOT PLATFORM FOR QUALITY, HEALTH AND MARKET DECISIONS IN TEA FARMING

Project ID: RP [25-26J-178]

Project Proposal Report

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DATA DRIVEN FERTLIZATION USING IoT DEVICE

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Declaration

Declaration page of the candidates & supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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29/08/2025

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Abstract

Tea is one of the globe's most important plantation crops, and productivity and quality are to a great extent influenced by efficient nutrient management. Fertilization in tea plantations has also been carried out in the past on a fixed calendar or experience-based basis, which tends to result in excessive or insufficient fertilizer application. The practice not only increases input expenses but also reduces nutrient use efficiency and encourages soil degradation, leaching of nutrients, and environmental pollution. In order to circumvent these limitations, this research recommends the development of an IoT-based data-driven tea plantation fertilization system. The system will consist of an array of sensors including soil nutrient probes nitrogen, phosphorus, potassium, soil pH and electrical conductivity sensors, soil moisture and temperature probes, and environmental sensors for monitoring rainfall, humidity, temperature, and solar radiation. Plant health indicators in terms of chlorophyll content and NDVI-based metrics will also be provided to monitor nutrient uptake and growth conditions. The data will be collected from the sensors and processed through a microcontroller-based system that will do real-time acquisition, minimal preprocessing, and wireless transmission to a cloud platform for analysis. Filtering, normalization, and data analysis using machine learning models will happen in the cloud platform. Both type-based rule limits and predictive models will be utilized to determine fertilizer requirements by type, dosage, and application time. The pilot field trial will be conducted by dividing a tea estate into two groups. One with the conventional farmer-managed fertilization and the other with the support of the IoT-based decision support system. Performance will be measured by comparing yield, biochemical quality of leaves, fertilizer-use efficiency, and soil health indicators. The predicted outcomes of the project include reductions in fertilizer consumption and cost, improved tea yield and quality, and more sustainable soils. The system is designed to provide a low-cost, scalable, and feasible model for precision agriculture in tea production that offers economic value to farmers and environmental advantages to long-term plantation stewardship.

Keywords – *Fertilization, normalization, analysis, machine learning models, cloud platform*

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Abbreviations

EC – Electrical Conductivity

IoT – Internet of Things

ML – Machine Learning

NDVI – Normalized Difference Vegetation Index

NPK – Nitrogen, Phosphorus, and Potassium (primary nutrients)

1. Introduction

Tea (Camellia sinensis) is among the world's most widely consumed beverages, with huge economic and cultural importance in producing countries such as India, Sri Lanka, Kenya, and China. The tea industry supports millions of livelihoods through the cultivation, processing, and trading of tea. However, tea yield and quality are very much dependent on proper soil fertility management. Fertilizers play a crucial part in providing major nutrients such as nitrogen (N), phosphorus (P), and potassium (K) that directly affect leaf yield along with quality parameters such as theaflavins, catechins, and amino acids [1].

Traditional fertilization in tea plantations is often scheduled or experience-based rather than determined by real-time plant and soil requirements. This results in inefficient nutrient utilization, soil degradation, and increased production costs [2]. In the face of increasing concerns over sustainability and resource use efficiency, the application of precision agriculture technologies such as the Internet of Things (IoT) offers an opportunity to streamline fertilizer application, increase yields, and maintain long-term soil fertility.

1.1. Nutrient Management Challenges in Tea Cultivation

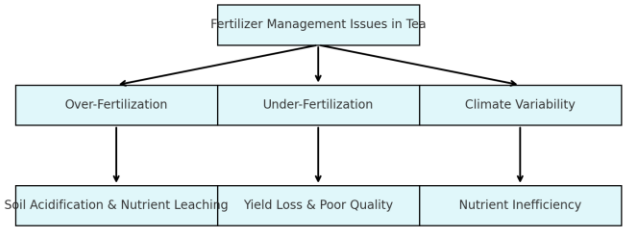


Figure 1: Nutrients management problems in Sri Lanka

1.1.1. Over-Fertilization and Soil Degradation

Excessive use of chemical fertilizers is a widespread practice in intensive tea production. Farmers use more doses in order to gain more yields, but the excess results in soil acidification, leaching of nutrients, and contamination of groundwater [3]. In acidic soils of Assam and Sri Lanka, the practice hastens soil loss and diminishes the long-term sustainability of tea estates.

1.1.2. Under-Fertilization and Yield Losses

Conversely, in the majority of smallholder-managed tea farms, the scarcity of resources leads to under-application of fertilizers. This reduces the levels of available nutrients, which stunts plant growth, decreases leaf yield, and deteriorates quality [4]. Such practices compromise both profitability and competitiveness in the global tea markets.

1.1.3. Climate Variability

Tea nutrient acquisition is extremely climate-sensitive, specifically precipitation and temperature. Heavy rainfall can leach down applied fertilizers before their uptake by the plants, while drought limits nutrient mobility in the soil [5]. The conventional practices of fertilization do not consider these dynamic changes, contributing to inefficiency.

1.2. Role of IoT in Precision Fertilization

1.2.1. Sensor Integration

IoT technology makes it possible to monitor soil, plants and environmental factors in real time using sensors. Soil nutrient sensors (NPK, pH, electrical conductivity), soil moisture and temperature sensors, and environmental sensors (rainfall, humidity, solar radiation) provide valuable information about the field conditions. Plant health monitoring equipment such as chlorophyll meters (SPAD) and NDVI imaging also give real-time information on nutrient status [6].

1.2.2. Microcontroller-Based Data Collection

The sensor nodes are interfaced with microcontrollers that preprocess and accumulate data prior to wirelessly sending it to cloud platforms. Microcontrollers have widespread use in IoT due to their low cost, energy efficiency, and presence of wireless connectivity features [7]. This enables the scalable and affordable deployment of smart farming methods in tea gardens.

1.2.3. Data-Driven Decision Support

After being carried by the data, it is analyzed through algorithms and machine learning models to give fertilizer recommendations. These models can predict nutrient needs, recommend appropriate dosages, and schedule application dates based on weather conditions and soil types [8]. Decision support systems provide farmers with actionable data, allowing them to use the right fertilizer at the right time in the right amount.

2. Background

Tea (Camellia sinensis) is the most consumed drink globally and holds enormous economic, social, and cultural importance. It is cultivated in over 50 countries, with the largest proportion of world production coming from Asia and Africa. China, India, Sri Lanka, and Kenya are the dominant producers that cover the world market for tea and are also responsible for generating foreign exchange earnings and employment [1]. To the majority of rural communities, tea cultivation brings not only revenue but also social stability and therefore is an essential agricultural commodity.

Soil fertility and nutrient management play a significant role in influencing tea yield and quality. The fertilizer application, particularly nitrogen (N), phosphorus (P), and potassium (K), directly impacts the yield of leaves in tea, the chemical composition of harvested shoots, and the sensory quality of processed tea [2]. It has been shown by research that optimally balanced fertilizing significantly improves theaflavins, catechins, and amino acid content, which are major determinant factors for flavor and market value in tea [9]. The application of nutrients in an unbalanced manner can reduce leaf quality, yield potential, and long-term soil fertility.

Fertilization management in the majority of tea production regions is still in-effective even though it is important. Conventional methods commonly follow blanket application of fertilizers at fixed frequencies regardless of real-time plant or soil condition. Over-fertilization is widespread in intensively cultivated tea estates, leading to acidification of soil, loss of nutrients by leaching, and environmental problems such as contamination of groundwater [3]. Under-fertilization occurs with resource-poor smallholder producers, leading to reduced yields and poor leaf quality [4]. In both situations, nutrient use efficiency is not good, and cost of production is unnecessarily raised.

Climate variability compounds the complexity of nutrient management. Rainfall, temperature, and humidity control nutrient uptake and availability. For instance, heavy rain can leach applied fertilizers before roots of the plants absorb them, whereas extended dry

periods lower nutrient solubility and mobility [5]. As conventional fertilization methods hardly change in response to these fluctuating conditions, tea estates tend to be confronted with variable yields and irregular quality. With climate change also posing additional challenges to tea production, adaptive and information-based solutions are the order of the day [10].

Precision agriculture has emerged as an advanced way to optimize the utilization of resources while increasing sustainability in the face of such challenges. Precision agriculture involves site-specific management of crops with the help of real-time monitoring of the soil, plant, and environmental conditions [6]. The Internet of Things (IoT) is at the forefront of this revolution, which allows farmers to invest in sensors, monitor continuous field data, and make decisions based on advanced analytics [7]. IoT-based monitoring allows predictive estimation of nutrient requirements, optimization of fertilizer dosage rates, and timing of fertilization as per climatic conditions [8].

Though there has been a lot of research on IoT applications in agriculture on crops like rice, maize, and wheat, it is still less researched on perennial plantation crops like tea. For instance, IoT-based systems have been used to schedule irrigation, monitor greenhouses, and soil nutrient analysis for vegetables and cereals [11], but not much case study is done on tea or coffee plantations. Given the global importance of tea and the unique challenges of its cultivation, there is a clear possibility of using IoT-driven, data-based fertilization systems for tea estates. These systems can use soil nutrient sensors, climate sensors, and plant health sensors to make precise suggestions, thereby preventing wastage, promoting high yields, and ensuring eco-friendliness [12].

This research attempts to bridge that gap by designing and testing an IoT-based fertilization system for tea estates. Relying on real-time sensing, microcontroller-assisted data acquisition, and analytics facilitated by machine learning, the research tries to demonstrate how modern technology can transform tea cultivation into a more efficient, profitable, and sustainable practice.

3. Literature Review

3.1. Challenges in Fertilization and Nutrient Management in Tea

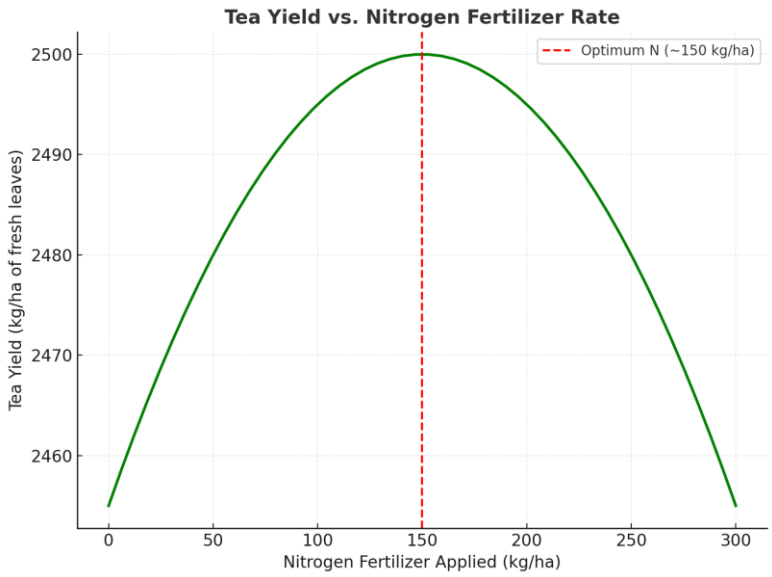


Figure 2: Tea yield vs Nitrogen fertilizer rate

Nutrient management in tea farming is complex because tea is a perennial crop with specific nutrient requirements and is typically grown under highly weather exposed conditions. Fertilizer application directly influences leaf yield, shoot growth, and the biochemical quality of the final product. However, after decades of agronomic research, tea farmers continue to face significant challenges to effective use of fertilizers. These can be grouped into agronomic, environmental, and technological factors.

3.1.1. Over-Fertilization and Soil Degradation

On highly managed estates, fertilizers are frequently used at high levels to optimize output. Although this may increase production in the short term, research has demonstrated that high levels of nitrogen and potassium application can cause acidification of the soil, nutrient loss through leaching, and reduced soil fertility [3]. Tea soils, such as Assam in

India and the Sri Lankan central highland, are naturally acidic, and over-fertilizing sweeps this trend, affecting microbial life and long-term sustainability [2]. Nutrient runoff during heavy rain also causes contamination of nearby water bodies, which can be harmful to ecosystems and human health [10].

3.1.2. Under-Fertilization and Yield Decline

Smallholder tea farmers, who are significant producers of tea globally, are not able to use optimum fertilizer application. Limited access to credit, unpredictable fertilizer costs, and weak extension services led to under-application of essential nutrients, thus lowering leaf yield and quality [4]. In Kenya, for instance, smallholder-managed tea plantations have substantial yield gaps compared to estate-managed plantations due to differences in the levels of nutrient application [5].

3.1.3. Climatic Variability and Nutrient Uptake

Tea is highly vulnerable to climatic factors such as rainfall, temperature, and humidity. Rainfall variability alone poses a huge challenge: pre-heavy-rain fertilizer applications leached prior to absorption, and prolonged dry spells reduce solubility of nutrients and root uptake [5]. Climate change-amplified increasing temperatures are projected to increase evapotranspiration, further exacerbating nutrient management [1]. The result is that conventional fixed-schedule fertilization is becoming increasingly inappropriate under changing climatic conditions.

3.1.4. Blanket Recommendations vs. Site-Specific Needs

Most of the fertilization practices in tea rely on broad guidelines developed years back, which ignore site-specific soil fertility, microclimate variation, and varietal response [2,9].

Tea fertilizer response varied extensively among regions as well as within estates depending upon soil, slope, and rain situation [3,5]. Poor fertilizer uses and loss of potential yield increase is the result of lacking site-specific nutrient management.

3.1.5. Limited Monitoring and Data-Driven Practices

As opposed to high-value crops such as greenhouse vegetables where sensors and precision tools are heavily used, tea farms do not typically use real-time monitoring systems for soil and crop health. Growers take subjective guesses based on visual observation of foliage color or plant condition to estimate nutrient status [7]. Fertilizer type, rate, and timing decisions are guesses without data, and varying results ensue.

3.1.6. Economic Constraints and Accessibility

Fertilizers account for a large proportion of tea production cost. In the case of smallholder farmers, rising input costs can force reduced application levels, which are part of the cause of nutrient deficiencies [4]. Further, in most tea-producing regions, access to newer technologies such as controlled-release fertilizers or biofertilizers is hindered due to their high cost and lack of awareness [10]. This is an economic barrier to the adoption of sustainable practices.

3.1.7. Lack of Integration with Modern Technologies

While precision agriculture and IoT technologies have been promising for cereals and horticultural crops [6,11], for tea their uptake is nil. Plantation managers do not know how such technologies can be optimized for perennials, while smallholders are constrained by cost and training [7]. Such non-integration also widens the gap between traditional and modern methods and makes tea plantations prone to inefficiency.

3.2. Critical Analysis of Work

3.2.1. Fertilizer Application

Nitrogen management studies in tea have both created opportunities and presented challenges. It has been reported that the use of controlled-release fertilizers has the effect of maximizing the efficiency of nitrogen utilization and reducing leaching in tea soil [10]. Similarly, in Kenya and Sri Lanka, research emphasized that fertilizer response is very site-specific, being dependent on soil type and microclimatic conditions [4,5]. This would mean that fixed fertilizer regimes do not optimize nutrient recovery.

3.2.2. Quality Parameters and Fertilizer Influence

Nutrient supply, besides influencing yield, also influences tea quality. Use of fertilizer improves the level of amino acids and quality of liquor [9]. However, imbalance in the use of fertilizers can influence quality, and there is therefore a need for accurate dosage and timing. Such evidence necessitates the integration of real-time monitoring inputs in the use of fertilizers.

3.2.3. Precision Agriculture and IoT

Precision agriculture has accelerated as a way to optimize input utilization. IoT technologies facilitate the real-time tracking of environmental and soil conditions, providing actionable suggestions for fertilizer scheduling [6]. Meta-analyses of IoT in agriculture define their potential for increased productivity with reduced wastage [7]. Big data analysis coupled with it further enhances predictive ability, facilitating adaptive decision-making in agricultural systems [11].

3.2.4. Machine Learning in Agriculture

Machine learning (ML) is picking up pace to support agricultural decision making. The uses range from yield prediction to nutrient management and pest identification [8,12]. For example, ensemble models and deep learning can more accurately predict nutrient requirements than usual statistical analysis. This provides compelling reason to implement ML in tea estates because of the complex soil–plant–climate interactions.

3.3. Emerging Trends in Smart Fertilization for Plantation Crops

Recent advancements in precision agriculture report various trends relevant to tea cultivation,

- IoT with Cloud Platforms - Low-cost sensors connected through microcontrollers enable real-time data acquisition and cloud storage, with farmers being able to make intelligent fertilizer decisions in real time [7,11].
- Predictive Analytics - Machine learning algorithms are being used more frequently to forecast nutrient needs and optimize fertilizer planning [12].
- Sustainable Fertilization Practices - Controlled-release fertilizers, organic amendments to matter, and sensor-based fertigation are being looked at as green alternatives [10].
- Application in Plantation Crops - While the use of IoT is strongly established for vegetables and cereals, usage in perennial plants like coffee and tea is limited, creating research and innovation opportunities [13].

4. Research Gap

- **Monitoring and Management**

A recent study of tea identifies remote sensing and mapping NDVI monitoring of tea condition, multi-source phenotyping, and deep-learning plantation mapping. But it does not fill the gap to bridge these data and bring them into prescriptive, per-block fertilization recommendations. In other words, we can see heterogeneity but few publications close the loop to dose timing conclusions for nutrients [15–17].

- **Nitrogen Focus Without Decision Systems**

Literature well evidence nitrogen uses inefficiency and loss (runoff, volatilization, N_2O/NO emissions) in tea and examines mitigation. But pragmatic decision-support systems that synthesize real-time soil, plant, and weather information to optimize N rate and timing are largely lacking. Field scale tools that co-optimize yield, quality, and environmental endpoints (e.g., N_2O emissions) are rarely illustrated [10,18].

- **Isolated Components, Little Integration**

There are parts available out their quality detection of new tea buds with N status correlated UAV platforms for precision agriculture. But they are component solutions. We have fewer end-to-end pipelines integrating in-field sensing to cloud analytics/ML and then actionable fertilizer set-points and finally evaluation spanning seasons. Multi-season, multi-block validation across real tea estates is sparse [15,16,19].

- **Perennial-Crop and Smallholder Gap**

Precision farming and IoT field surveys recognize adoption issues in terms of connectivity, cost, and management of data. Advanced case studies are predominantly on annual crops and glasshouses. Plantation perennial crops like tea are relatively less attended to especially in the guise of low-cost, microcontroller-based solutions suitable for smallholders [7,11,20,21].

- **Data and Benchmarking Limitations**

Cross-study comparison is hampered by non-standardized datasets, low-density open data on soil, plant, climate, management histories and a scarcity of tea nutrient recommendation task benchmarks. This retards model transferability and objective assessment. Overall smart-farming and big-data survey papers also bring out data governance and standardization as persistent barriers [11,14,22].

At the end of this discussion, this is the study will uniquely contribute,

An integrated IoT-enabled fertilization system for tea that combines soil, plant, and microclimate measurement with ML-informed fertilizer type, rate, and timing recommendations at block level. Field experimentation as a control against conventional practice for yield, quality, fertilizer-use efficiency increase, and reduced N emissions. A reusable data schema and evaluation protocol to facilitate increased comparability across studies and enable open datasets in the future for tea nutrient management.

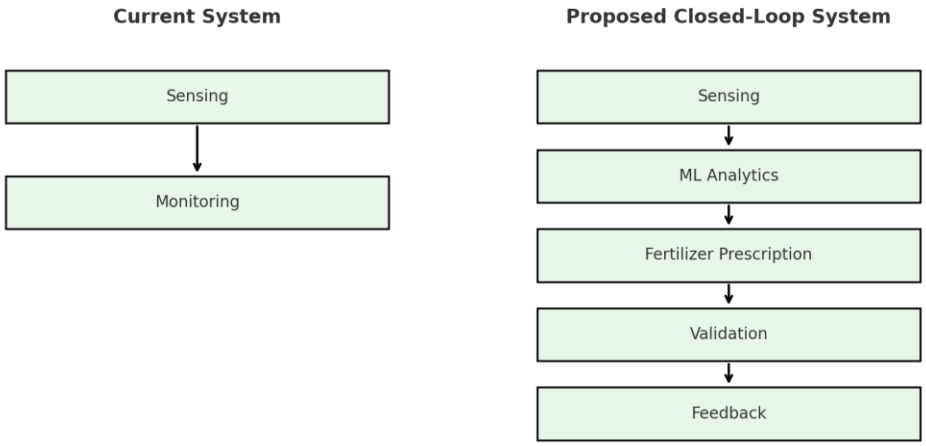


Figure 3: Current systems vs Proposed system

5. Research Problem

Tea is a globally significant plantation crop whose yield and quality are mainly in the hands of optimal nutrient management. Nevertheless, fertilization in tea plantations is still under the grip of conventional, fixed-frequency schedules that ignore site-specific heterogeneity of soils, momentary plant needs, and weather fluctuations [2,4,5]. As a result, tea cultivation is afflicted with two problems:

1. Inefficiency in Fertilizer Use

Overuse of fertilizers has led to acidification of soil, nutrient leaching and increased input costs, and inefficient application. This is common among poor smallholders, and it reduces yields and leaf quality [3,4,9].

2. Lack of Integrated Decision Support

Whereas IoT and machine learning technologies have been promising in precision agriculture for other crops [6,7,12,20,21], applications specific to tea do not exist. Earlier studies provide monitoring solutions like NDVI mapping, nutrient sensing, or measures of soil health. But these are solitary components and not integrated into actionable, closed-loop fertilization systems [15–19].

The main problem therefore is,

Tea states lack an integrated, IoT-based, machine learning–equipped fertilization system able to convert real-time soil, plant, and environmental data into actionable fertilizer prescriptions (type, rate, and timing) validated in field settings.

This problem creates ongoing inefficiency, cost increase, farmer profitability loss, and long-term environmental risks such as soil degradation and greenhouse gas emission [1,10,18].

6. Objectives

6.1. Main objective

To develop and pilot an IoT-enabled, data-intensive fertilization system for tea estates that integrates soil, plant, and environmental monitoring with machine learning–based decision support for optimizing fertilizer usage, yield improvement and quality, and promoting sustainable soil health management.

6.2. Specific objectives

- To design and implement an IoT-based monitoring system for tea estates. This includes the installation of soil nutrient sensors (NPK, pH, EC, moisture), environmental sensors (temperature, humidity, rainfall, solar radiation), and plant health indicators (chlorophyll and NDVI). The goal is to achieve a high-performance, microcontroller-based sensing system capable of performing continuous, real-time data acquisition in diverse tea-growing environments.

- To preprocess, process, and analyze the data acquired for decision-making. Raw sensor data will be cleaned, calibrated, and normalized to remove errors and noise. A cloud database will be established to store and merge multi-dimensional data (climate, soil, and plant). The objective is to make the pipeline data robust, scalable, and prepared for advanced analysis.

- For constructing and training machine learning models for fertilizer recommendation. Using the processed field data, machine learning algorithms (regression, ensemble methods, and possibly time-series modeling) will be trained to predict nutrient requirements. They will give site-specific fertilizer type, application rate, and timing advice, bridging the gap between monitoring and actionable farm operations.

- To conduct field trials comparing IoT-based fertilization with conventional practices. A controlled experiment will be done in selected tea plots, one of which will be run with farmer-practice fertilization and the other under guidelines based on IoT. The performance will be evaluated in terms of yield of tea leaves, biochemical quality (theaflavins, amino acids), fertilizer use efficiency, and soil health, providing a clear comparative analysis of the performance.

- For assessing scalability, cost-effectiveness, and sustainability of the proposed system. Beyond technical verification, this objective will evaluate economic sustainability among smallholders, adoption possibilities in large estates, and the long-term environmental benefits like reduced loss of soil and minimized nutrient loss. The objective is to demonstrate the practicality and broader implications of the system on the tea industry.

7. Methodology

7.1. Component Overview

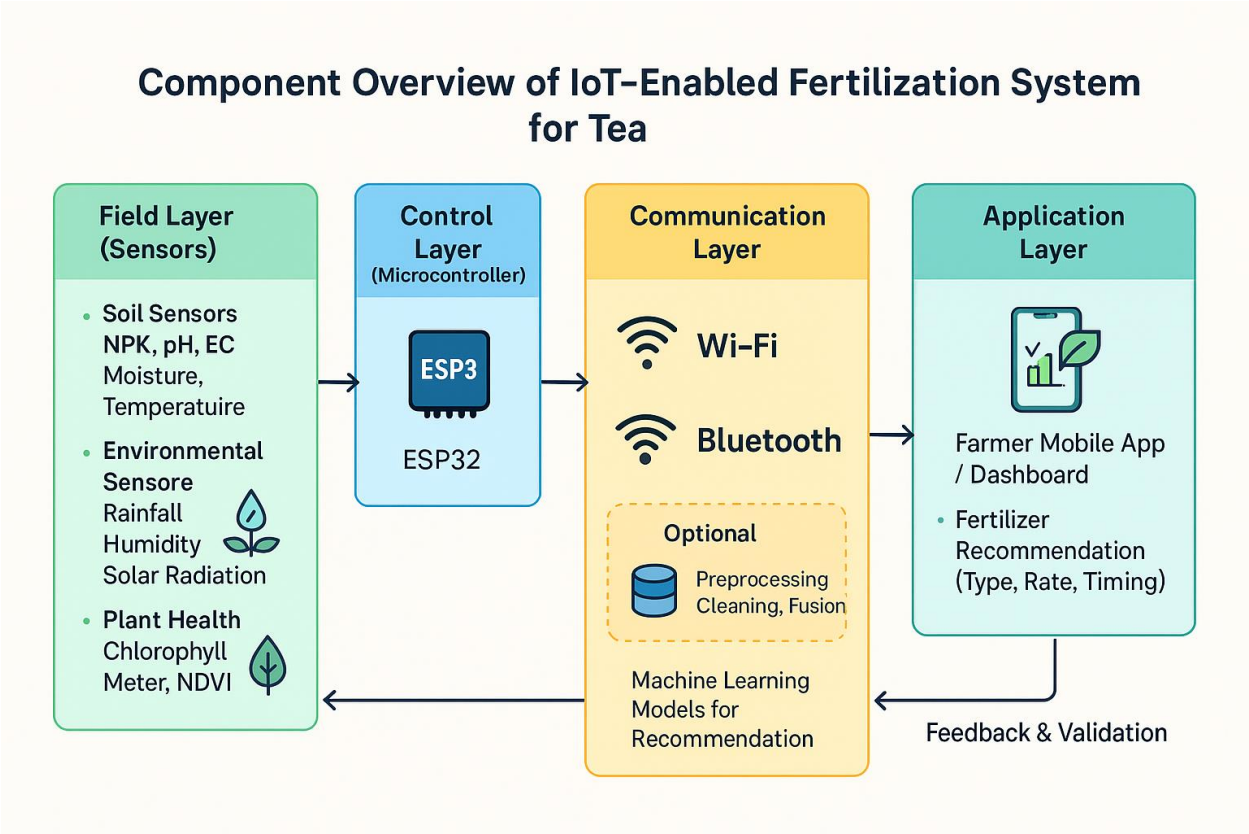


Figure 4: Component overview diagram

The component diagram shows the flow of the IoT-based fertilization system for tea plantations. At the field layer, different sensors continuously monitor soil parameters (NPK, pH, EC, moisture, temperature), environmental parameters (rainfall, humidity, solar radiation), and plant health metrics (chlorophyll, NDVI). The readings are transmitted to the ESP32 microcontroller, which is the local control unit by acquiring, preprocessing, and packaging the data for transmission. The communication layer connects, via Wi-Fi or Bluetooth for short-distance data transmission, with the alternative of LoRa or cellular networks for distant plantations. The data, once transmitted, reaches the cloud layer, where

it is securely stored, cleaned, and integrated. In this layer, machine learning algorithms sort through patterns and give fertilizer recommendations by computing the type, dosage, and timing needed for the prevailing field conditions. At the top, the application layer shows these recommendations to farmers through a mobile app or dashboard with simple and actionable advice. A feedback mechanism allows farmers to input responses, allowing the system to render its recommendations increasingly accurate and improved over time.

7.2. Methodology used

The research method focuses on the design and field testing of an IoT-based data-driven fertilization system for tea farms. The methodology consists of five main phases: system design, data collection, processing of data, implementation of the model, and field validation.

7.2.1. System Design and Integration of Components

- An IoT-capable monitoring platform will be developed with soil, environmental, and plant health sensors.
- These sensors will also capture NPK, pH, EC, soil moisture, temperature, rain, humidity, and chlorophyll content levels.
- The sensors will be interfaced with an ESP32 microcontroller, being the control unit, which will be tasked with data acquisition and primary preprocessing.
- Data will be transmitted via Wi-Fi or Bluetooth with LoRa or cellular networks utilized in far-flung plantations.
- There will be an online platform for storing, preprocessing, and integrating multi-source data.

7.2.2. Data Acquisition

- Data will be collected on a real-time basis from tea plantation plots installed with the IoT platform.
- Soil and plant sensors will provide nutrient-related values, whereas environmental variability will be recorded by climate sensors.
- Data will be timestamped (if required), geotagged, and automatically uploaded to the cloud.
- Fertilizer application records maintained by farmers will also be included for comparison.

7.2.3. Data Processing and Management

- Data collected will undergo preprocessing activities such as calibration, noise filtering, and normalization.
- Data fusion algorithms will combine soil, climate, and plant health information to create a holistic dataset.
- Outlier detection techniques will be applied to remove erroneous readings.
- Clean data will be stored in a structured database for analysis.

7.2.4. Machine Learning Model Development

- Machine learning models (e.g., regression models, ensemble algorithms, or time-series forecasting) will be trained on the cleaned dataset.
- These models will forecast nutrient requirements and generate fertilizer advice (type, dosage, and timing).
- Model performance will be verified with metrics such as accuracy, RMSE (Root Mean Square Error), and prediction reliability.
- A decision-support dashboard will be designed to give farmers recommendations.

7.2.5. Field Validation Experiment

- A pilot experiment will be carried out in a selected tea plantation plot.
- Two treatments will be compared:
  - ❖ Conventional fertilization practices (fixed schedule, farmer-managed).
  - ❖ IoT-based, ML-driven fertilization (system-generated recommendations).
- Key parameters will be monitored:
  - ❖ Yield (fresh tea leaves kg per hectare)
  - ❖ Leaf quality (theaflavins, amino acids, catechins)
  - ❖ Use efficiency of fertilizers (FUE%).
  - ❖ Soil health parameters (stability of pH, organic matter).
- The experiment will be conducted over a few cycles of plucking to account for seasonal changes.

7.2.6. Evaluation and Feedback

- The results of both the treatments will be statistically compared to determine the effectiveness of the IoT-based system.
- Perceptions of farmers regarding usability, cost, and feasibility of the system will be collected.
- There will be a feedback loop included, allowing for continuous tuning of the ML models and system recommendations.

7.3. Technologies

To render the proposed IoT-based fertilization system viable and cost-effective for the Sri Lankan tea industry, cost-effective technologies are selected at every layer of the system.

At the field layer, low-cost farm sensors will be utilized for tracking soil, climate, and plant status. Soil condition will be tracked with a combination of low-cost probes,

including a DFRobot analog pH sensor and a simple EC (electrical conductivity) circuit for salinity measurement. Capacitive rather than resistive sensors will be used, since the former are more durable and provide more accuracy in field environments. Soil temperature will be sensed using a DS18B20 water-resistant temperature sensor, while air temperature and humidity will be sensed using compact digital sensors such as the SHT31 or, where budgets are more limited, the DHT22. Rain can be measured by a tipping-bucket rain gauge, and ambient light conditions will be recorded by a BH1750 light sensor as an inexpensive surrogate for solar radiation. Intermittently, handheld chlorophyll meters (SPAD clones) or phone-based NDVI measurement tools can be used in lieu of expensive image-taking equipment.

For the measurement of soil nutrients (N, P, K), the system will utilize inexpensive ion-selective electrode (ISE) technology-based soil nutrient sensor modules to provide approximate measurements of the nitrogen, phosphorus, and potassium concentrations in the soil solution. While less precise than laboratory-quality instruments, these sensors will provide useful trend data if calibrated to ground-truth results. For reliability improvement, other ancillary parameters such as soil pH, EC, and plant chlorophyll content will be utilized as indirect measures of nutrient status. Furthermore, leaf and soil samples will be taken at regular intervals in the laboratory with standard agronomic protocols and utilized for machine learning model calibration and NPK sensor calibration. By combining direct sensor data, proxy parameters, and laboratory checks, the system will offer improved reliable nutrient recommendations at low prices.

In the control layer, the ESP32 microcontroller gives the best balance between cost and ability. It supports Wi-Fi and Bluetooth communication, can take multiple sensors' inputs, and is easily available in Sri Lanka at low prices. Where analog sensors like pH and EC are used, an ADS1115 analog-to-digital converter can be added for improved precision. Buffering data locally can be supported by a low-cost microSD module. The field equipment will be powered with small solar panels (6–10 W), supported by low-cost TP4056 recharging modules and rechargeable 18650 Li-ion batteries. All the electronics

will be in weather-resistant IP65 enclosures with cable glands and silica desiccants to protect them from moisture damage.

The communications layer will predominantly be Wi-Fi where there is plantation connectivity because it lowers cost and complexity. In plots which are remotely located relative to Wi-Fi connectivity, ESP32 modules can temporarily use Bluetooth for configuration or data transfer. Where necessary, low-cost LoRa modules (SX1276/78) can be added to ensure additional range, and one LoRa gateway can harvest data from numerous field nodes. In extremely remote areas, a minimal 4G USB dongle plugged into a Raspberry Pi gateway can serve as the backhaul to transport data to the cloud.

Data management and the cloud layer will be built entirely using free, open-source products. Data would be consumed via a tiny MQTT broker such as Mosquitto and pre-processed with Node-RED flows. InfluxDB will be used as a time-series database to store data in the long term, and Grafana will provide an easy but efficient interface to dashboards and visualizations. API services may be hosted on a minimalist Python-based framework such as Flask or FastAPI if necessary.

For the analytics layer, machine learning models will be developed using open-source Python packages. Basic threshold-based rules can be used for early decision-making, and the optimization thereafter can be performed using algorithms such as Random Forest or XGBoost for predicting fertilizer dose and timing. Forecasting nutrient needs on a seasonal basis can be developed with lightweight tools such as Prophet. All the processing will be performed on a central server or cloud for instance, leaving the computational load away from low-power field devices.

Finally, the application layer will be tasked with displaying results to farmers and plantation managers in a comprehensible and simple manner. On the least cost grade, real-time graphs and recommendations can be provided through Grafana dashboards, with notifications sent through SMS, WhatsApp, or email using free webhook integrations. For added convenience, a simple web application can be designed employing open-source systems with Sinhala, Tamil, and English interfaces. The farmer can be notified of the

fertilizer type, amount, and application time as well as warning messages for unfavorable weather conditions for fertilizer application.

For the upkeep of accuracy, low-cost calibration standards such as pH buffer solutions, EC reference solutions, and standard nutrient calibration kits will be used in order to keep sensors precise. Periodic leaf or soil testing in the laboratory will serve as a ground-truthing technique for checking data as well as machine learning recommendations. Utilizing low-cost hardware, free open-source software, and simple power sources, this setup is an affordable yet scalable technology stack that can be feasibly implemented in Sri Lankan tea plantations.

8. Requirements

Functional requirements dictate what the system must do to meet its purpose. For this research study, the main functional requirements are:

8.1. Functional Requirements

1. Sensor Data Acquisition
   - The system must collect real-time data on soil measurements (pH, EC, moisture, temperature, NPK levels), environmental aspects (rainfall, temperature, humidity, light intensity), and plant health (chlorophyll/NDVI).

2. Data Transmission and Storage
   - The system will have to transfer field data to the cloud over Wi-Fi, Bluetooth, LoRa, or 4G, based on availability.
   - Data will have to be stored in a structured database for long-term use and analysis.

3. Data Preprocessing
   - The system will have to pre-process data to clean, filter, and normalize the data for noise and error removal.
   - Sensor accuracy needs to be ensured through the application of calibration functions.

4. Machine Learning Based Analysis
   - The system will have to analyze aggregated soil, climate, and plant health data using ML models.
   - The ML model must generate fertilizer recommendations in terms of type, quantity, and schedule.

5. User Interface and Decision Support
    - Farmers must be provided with user-friendly dashboards or interfaces on mobile applications.
    - Recommendations must be communicated in Sinhala, Tamil, and English with explicit actionable instructions.
    - Users must be reminded through alerts when it is inappropriate to use fertilizers.
6. Feedback Loop
    - The system must enable farmers to provide feedback on action taken.
    - Feedback data must be used to improve models and enhance accuracy over time.

8.2. Non- Functional Requirements

These define the system's characteristics and constraints beyond its functionality.

01. Reliability
    - The system must be >90% available in normal field deployment.
    - Data transfer must buffer to avoid losses when networks fail.

02. Scalability
    - The system must be able to scale from tens of nodes (prototype) to hundreds or dozens of nodes (plantation-scale).

03. Affordability
    - The system must utilize low-cost sensors, ESP32 microcontrollers, and open-source software so that it remains within the affordability of Sri Lankan tea farmers.

04. Energy Efficiency

- Field nodes must be powered by small solar panels (6–10 W) with backup batteries and be able to function reliably without grid power.

05. Maintainability

- Easy calibration of sensors (pH buffer, EC standards) and modular replacement of broken components should be facilitated in the system.

06. Security

- Data transmission must be done over secure protocols (MQTT over TLS/HTTPS).
- Each device must possess its own authentication credentials.

07. Usability

- The farmer-user interface must be simple, mobile-device-compatible, and require few technical specifications.

8.3. User Requirements

User requirements define the expectations and requirements of the end-users (farmers, plantation managers, and researchers).

1. Farmers

- Receive easy fertilizer guidance (what, how much, when to apply).
- Receive a reminder not to fertilize in heavy rain or unsuitable conditions.
- Access the system in Sinhala/Tamil, with graphical depiction for easier understanding.

2. Plantation Managers

- Have the ability to monitor multiple fields/blocks in a centralized dashboard.
- Compare yield and input efficiency between traditional and IoT-optimized plots.
- Download reports to support decision-making and record-keeping.

3. Researchers/Technicians

- Use raw sensor and process data for analysis and model refinement.
- Do calibration and validation through the inclusion of lab results.
- Test and optimize new algorithms or fertilizer recommendation schemes.

9. Commercialization

The proposed IoT-driven fertilization system for tea estates is strongly commercially viable, directly addressing among the most crucial issues both smallholder farmers and large plantation estates in Sri Lanka currently grapple with. Rising fertilizer costs, soil fertility reductions, and the calls for sustainable agricultural practices at long last caused a palpable market demand for cost-effective precision agriculture solutions. By offering a platform that synergizes real-time environmental, soil, and plant monitoring and machine learning–based fertilizer guidance, the system benefits various stakeholders including plantation managers, farmers, researchers, and policymakers.

A phasing strategy of commercialization is proposed to tackle sustainable growth and adoption on a scale.

Phase 1: Prototype Deployment and Pilot Trials

- Early deployment in collaboration with the Tea Research Institute (TRI) and selected smallholder samples in the Central Highlands.
- Demonstrate proof-of-concept by comparing IoT-guided fertilization with standard farmer practice.
- Obtain feedback from farmers and plantation managers to make iterations on system design and interfaces.
- Increase awareness through demos, workshops, and publications on cost savings and yield improvement.

Phase 2: Smallholder Farmer Adoption

- Develop a low-cost subscription model in which farmers get recommendations through a mobile application or SMS alert.
- Establish partnerships with farmer cooperatives, plantation companies by region, and agribusiness services providers for marketing the technology.

- Provide training and local-language support to enable uptake among non-technical farmers.
- Offer flexible pricing, e.g., low-cost base package with optional extra advanced analytics, to drive uptake among resource-poor smallholders.

Phase 3: Enterprise Licensing for Large Plantations

- Customize the platform for integration with estate-scale management systems, to enable block-level nutrient management and yield optimization.
- Provide enterprise dashboards, advanced analytics, and multi-plot monitoring tools for plantation managers.
- Develop regional plantation firm, export-oriented tea manufacturing, and sustainability certification body license models and long-term service agreements.
- Highlight environmental benefits like reduced nutrient runoff and improved soil health, helping estates meet international levels of sustainability (e.g., Rainforest Alliance, Fair Trade).

Phase 4: Scaling and Global Expansion

- Scale the system to other big tea-producing countries like India, Kenya, and China.
- Scale up the system to other perennials (coffee, coconut, rubber) to expand the addressable market.
- Develop strategic alliances with international agricultural research centers, agritech entrepreneurship, and governmental agencies.
- Customize a range of linguistic and cultural environments to obtain maximum global uptake.

10. Description of Personal and Facilities

Personnel

The project team has an alliance of academic guidance, industry collaboration, and field-level expertise to attain effective implementation.

- Principal Researcher (Student) - Responsible for overall project realization, including system design, data collection, analysis, and reporting.
- Supervisor and Co-Supervisor (University Lecturers) - Provide academic guidance, research methodology assistance, and technical experience in IoT applications, agriculture, and data analysis.
- Dilmah Tea Staff (Head Office) - Offer logistical support, coordination for estate management, and plantation data access related to fertilizer management.
- Kahawaththa Estate Farmers and Field Staff - Assist field trials with plot management, helping in sensor installation, and providing experiential knowledge on prevailing fertilization practices.

Having this combination of academic and industry staff assures us that the project is scientifically valid and practically relevant to the tea industry.

Facilities

The project will leverage both industry and university facilities to enable research work

- Dilmah Tea – Kahawaththa Plantation - The primary field site for prototype deployment and pilot trials. The plantation provides access to tea fields, farmers' expertise, and estate-level fertilization regimes, to enable experimentation of the IoT system in real-world situations.

- Dilmah Tea Office Facilities - Provides administrative and technical support, e.g., access to plantation records, agronomic data, and liaison with field staff.
- University Facilities - Use of the faculty expertise, research mentorship, and soil and leaf sample testing laboratory facilities. University computer facilities will also be used for initial data analysis and simulation.
- IoT Computing Platforms and Hardware - Project-specific hardware like ESP32 microcontrollers, environmental and soil sensors, and cloud storage/analytics platforms will be used for system development and testing.

11. Budget and Budget Justification

Table 1: Budget

Item	Unit Price (LKR)	Qty	Total (LKR)
Sensors & Field Modules			
3-in-1 NPK (or multi-probe) Soil Sensor	23,890	1	23,890
Capacitive Soil Moisture Sensor	290	3	870
ESP32 Dev Board	1,490	3	4,470
pH Sensor with Probe	3,950	1	3,950
Handheld SPAD Chlorophyll Meter (clone)	7,000	1	7,000
Power & Enclosures			
Solar Panel 10 W + Charger + 18650 battery	~7,000	2	14,000
Waterproof Enclosure + Cable Glands	~3,000	1	3,000
Cloud & Data Infrastructure			
Cloud VPS (MQTT, InfluxDB, Grafana, etc.)	~5,000 annually	1	5,000
Total Estimated Budget			**58,180**

11.1. Justification

- 3-in-1 NPK Soil Sensor (23,890 LKR) - A NPK, pH, EC, moisture, and temperature multi-parameter sensor that keeps hardware complexity and cost to a minimum.
- Capacitive Soil Moisture Sensors (290 LKR per unit) - Inexpensive, rugged, and corrosion-resistant sensors suitable for field use.

- ESP32 Dev Boards (1,490 LKR per board) - Conveniently available, low-cost microcontrollers with built-in Wi-Fi/Bluetooth capability, suitable for prototyping.
- pH Sensor Probes (3,950 LKR) - A relatively low-cost probe for accurate soil acidity measurement, essential to soil health analysis.
- Solar Power + Storage (14,000 LKR total) - Adds off-grid capability with reliability. Includes solar panels, battery, and charging circuit.
- Protective Enclosures (6,000 LKR total) - Weather-resistant Enclosures to protect electronics from water and dust.
- Cloud Services (5,000 LKR annually) - Covers server charges data intake, processing, and dashboard hosting using open-source software like Mosquitto, InfluxDB, and Grafana.

12. Research Planning and Timeline

Project Initiation (1 month)

- Define project scope, goals, and expected outcomes.
- Determine stakeholders (Dilmah Tea, Kahawaththa estate farmers, supervisors, university).
- Create project infrastructure (project repo, comms channels, cloud setup).
- Order initial hardware (ESP32 boards, sensors, solar kits, SPAD meter).

System Setup & Data Collection (1.5 months)

- Install IoT devices (soil, climate, plant health sensors) at Kahawaththa estate pilot plots.
- Configure ESP32 microcontrollers for data collection and transmission.
- Install Wi-Fi / LoRa / 4G connectivity and solar-powered enclosures.
- Collect raw data (rainfall, soil NPK, pH, EC, moisture, temperature, chlorophyll/SPAD, humidity).
- Record farmer fertilizer practices as a baseline for future comparison.

Data Preprocessing & Integration (1 month)

- Preprocess sensor data (calibration offsets, noise filtering).
- Integrate multi-source data (climate + soil + plant health).
- Normalize datasets to feed machine learning models.
- Implement database pipelines (MQTT → InfluxDB / PostgreSQL).

Machine Learning Model Development (2.5 months)

- Develop rule-based baseline suggestions (threshold levels for NPK, pH, water).
- Train ML algorithms (Random Forest, XGBoost, or regression models) for fertilizer type–rate–timing prediction.

- Use time-series models (e.g., Prophet) for prediction of nutrient requirements.
- Integrate plant health (SPAD/NDVI) into nutrition recommendation reasoning.
- Build decision-support pipeline for real-time recommendation.

System Integration & Prototype Application (1.5 months)

- Develop mobile/web dashboard for farmers (Sinhala/Tamil/English).
- Integrate real-time recommendations into app (fertilizer type, rate, timing).
- Create weather-based advisory alerts (e.g., delay fertilization in case of heavy rain forecasted).
- Usability testing with Kahawaththa estate farmers initially.

Field Validation & Testing (2 months)

- Side-by-side testing
  - ❖ Control plots: conventional farmer fertilization.
  - ❖ Test plots: IoT + ML-guided fertilization.
- Track major parameters - yield, leaf quality (SPAD, theaflavins), fertilizer use efficiency, soil health.
- Test system accuracy, reliability, and acceptance by farmers.
- Collect improvement input from farmers/managers.

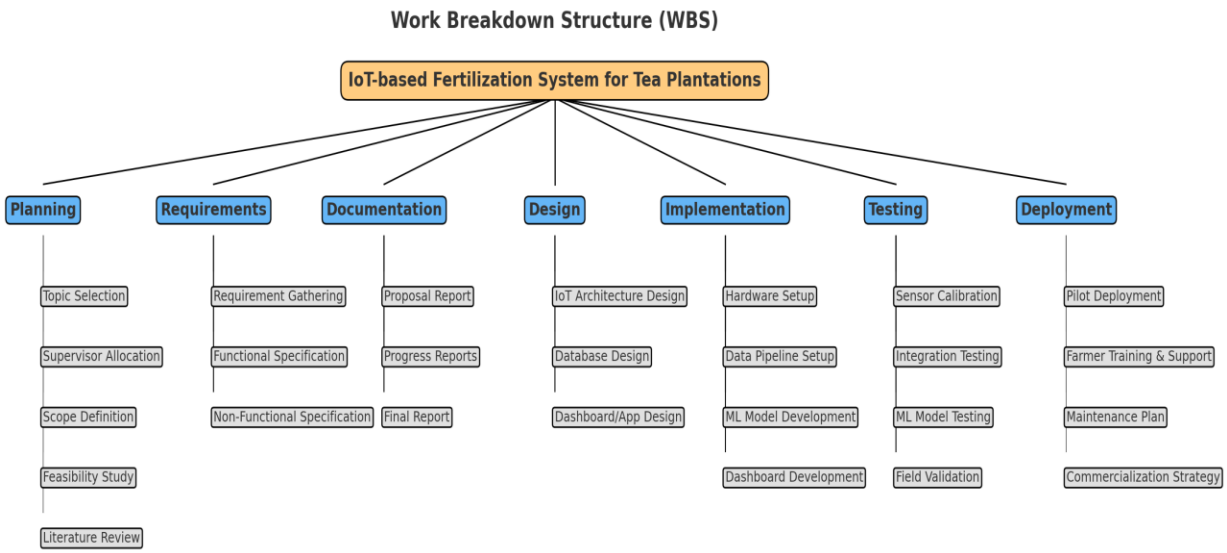
Optimization & Refinement (1 month)

- Optimization of sensor placement, power management, communication stability.
- Tune ML model parameters for increased accuracy and resilience.
- Improve app/dashboard user experience based on user feedback.
- Document operation issues and recommend solutions.

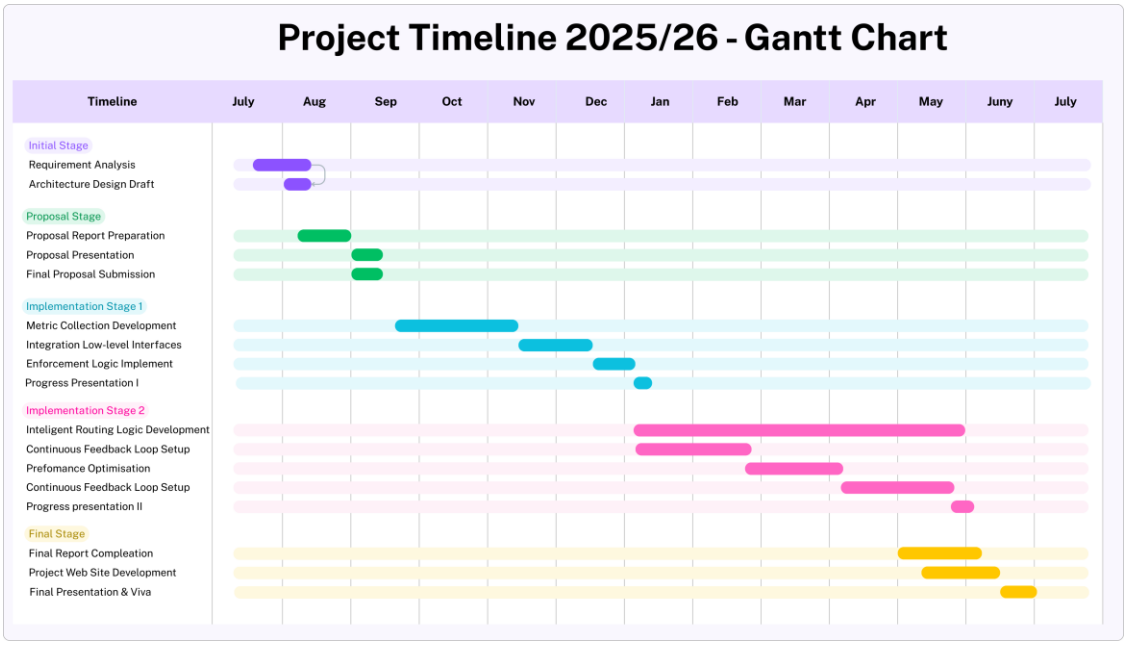
Documentation & Reporting (1 month)

- Prepare detailed technical documentation of system design and methodology.
- Construct datasets, analysis results, and validation outcomes.
- Prepare final thesis/report with commercialization plan.
- Report outcomes to supervisors, Dilmah Tea stakeholders, and academic readership.

12.1. Work Breakdown Structure



12.2. Gantt Chart



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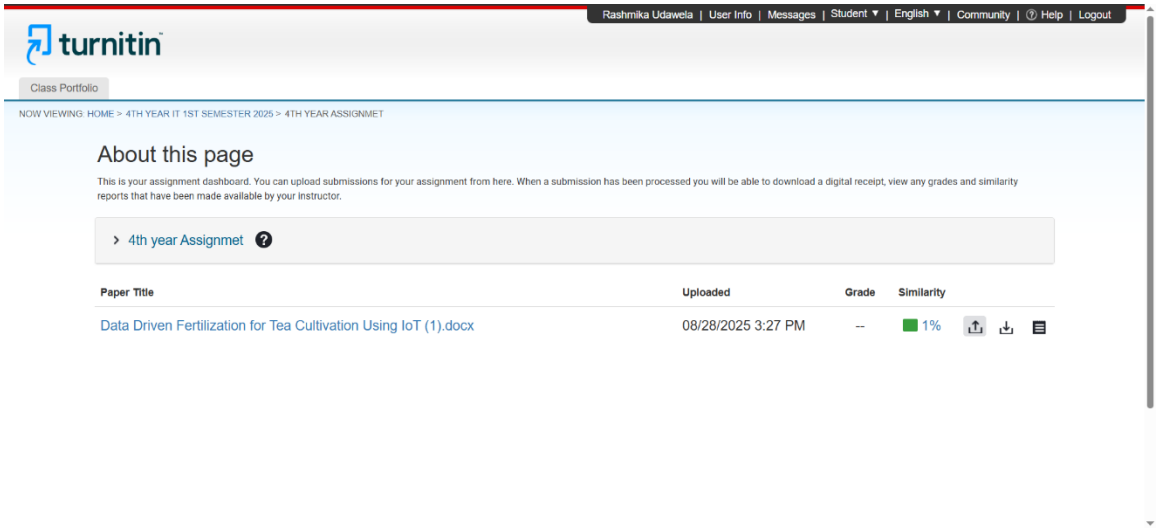
14. Appendix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